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# Enterprise Firewall Telemetry & Security Operations Center

## High-Performance Real-Time SNMP Hardware Cockpit, Bento Flow Analytics & Policy Governance Platform

**Document Reference:** CSIR-SERC/ICTD/EFTC/2026/DOC-01
**Classification:** Official Institutional Technical Manual
**Release Version:** 1.1.0 (InfluxDB 2.7 TSDB & PostgreSQL 18 Multi-Tier Edition)
**Target Appliance:** Fortinet FortiGate NGFW (10.10.3.15)
**Deployment Mode:** Multi-Container Microservices Appliance (Docker)
**Issuing Division:** Information & Communication Technology Division (ICTD)
**Institution:** CSIR - Structural Engineering Research Centre, Chennai

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# 1. System Architecture & Ingestion Pipeline

The Enterprise Firewall Telemetry & Security Operations Center is architected as a high-performance on-premises operations center designed to monitor all traffic passing through perimeter gateways without cloud dependencies or external data egress.

## Core Components

- **Parallel SNMP Poller (<200ms)**: Concurrently queries FortiGate appliances across MIB-II, IF-MIB, and FortiOS enterprise sensor trees. Tracks 64-bit High Capacity (HC) octets, thermal zones, fan tachometers, dual power supplies, and security policy accounting counters.
- **SNMP Trap Receiver (UDP 162)**: Native asynchronous UDP trap daemon listening on port 162, supporting both SNMPv2c (community: `fortinet`) and SNMPv3 (user: `fortinet` noAuthNoPriv). Ingests real-time hardware status changes, interface link up/down transitions, and FortiOS security alerts.
- **Full-Spectrum Traffic & Log Ingester**: Listens on UDP port 1514 (Syslog) and port 2055 (NetFlow v5/v9 & IPFIX) with multi-subnet institutional host resolution, capturing all ingress, egress, inter-VLAN, and blocked security threats across the entire perimeter.
- **InfluxDB 2.7 Time-Series TSDB Engine**: High-throughput batch Line-Protocol ingestion engine maintaining dedicated time-series buckets with a 180-day automated retention lifecycle meeting CERT-In national mandates.
- **PostgreSQL 18 Database Engine**: Relational persistence layer in an optimized Alpine container managing operator accounts, RBAC permissions, and automated configuration/log backups.
- **Reactive Vue 3.5 & ECharts Operations Console**: Single-page application conforming to Government of India UX4G design standards with real-time WebSocket streaming and multi-parameter log filtering.

## Network Port Matrix

| Port | Protocol | Direction | Source | Destination | Service Description |
|---|---|---|---|---|---|
| 80 / 443 | TCP | Inbound | Admin Workstation | Telemetry Console | Web UI & REST / WebSocket API |
| 161 | UDP | Outbound | Telemetry Console | FortiGate Gateway | SNMPv2c / SNMPv3 Polling |
| 162 | UDP | Inbound | FortiGate Gateway | Telemetry Console | SNMPv2c / SNMPv3 Trap Receiver (fortinet) |
| 1514 / 514 | UDP | Inbound | FortiGate Gateway | Telemetry Console | Hardware Syslog Forwarding |
| 2055 | UDP | Inbound | FortiGate Gateway | Telemetry Console | NetFlow v5/v9 & IPFIX Records |
| 8086 | TCP | Internal | Web Console | InfluxDB 2.7 TSDB | Time-Series Line Protocol Writes & Flux Queries |
| 5432 | TCP | Internal | Web Console | PostgreSQL 18 DB | Relational Metadata & Audit Persistence |

## 1.2 Universal Linux Automated Installer & Whitelabeling Wizard

The platform includes an automated universal installer script (`install.sh`) engineered for rapid production provisioning across **Ubuntu (20.04/22.04/24.04/Debian)** and **Rocky Linux (8/9/RHEL/AlmaLinux)** virtual machines and bare-metal servers:

> **One-Line Automated Deployment Command**
>
> ```
> curl -fsSL https://raw.githubusercontent.com/ananthkrishnangv/Enterprise-Firewall-Telemetry-Centre/main/install.sh | sudo bash
> ```

- **OS Detection & Security Updates**: Automatically detects Linux distribution, applies upstream operating system security updates via `apt-get` or `dnf`, and configures required base utilities (`curl`, `git`, `jq`, `openssl`, `net-snmp-utils`).
- **Node.js 22 LTS Runtime**: Installs current LTS Node.js and npm via official NodeSource enterprise repositories.
- **Container Engine & Compose**: Provisions official Docker CE engine and registers systemd daemon services.
- **Perimeter Host Firewall Configuration**: Automatically programs `ufw` (Ubuntu) or `firewalld` (Rocky Linux) rules for web (80/443), SNMP trap (162/udp), Syslog (1514/514/udp), and NetFlow (2055/udp).
- **Automated Database & TSDB Ingestion**: Launches isolated `postgres:18-alpine` and `influxdb:2.7-alpine` containers using a universal host-bridge Docker Compose configuration.
- **Interactive Terminal Whitelabeling Wizard**: Directly prompts the administrator for customized organization name, division, portal title, subtitle, vector logo preset, primary firewall IP, vendor, and SNMP community/v3 credentials.
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# 2. Visual Interface Tour

## 2.1 Authentication & First-Time Setup Wizard

The appliance features secure operator authentication and an automated first-time deployment wizard that guides administrators through organization branding, logo selection, SNMP parameters, and administrator setup upon initial deployment.

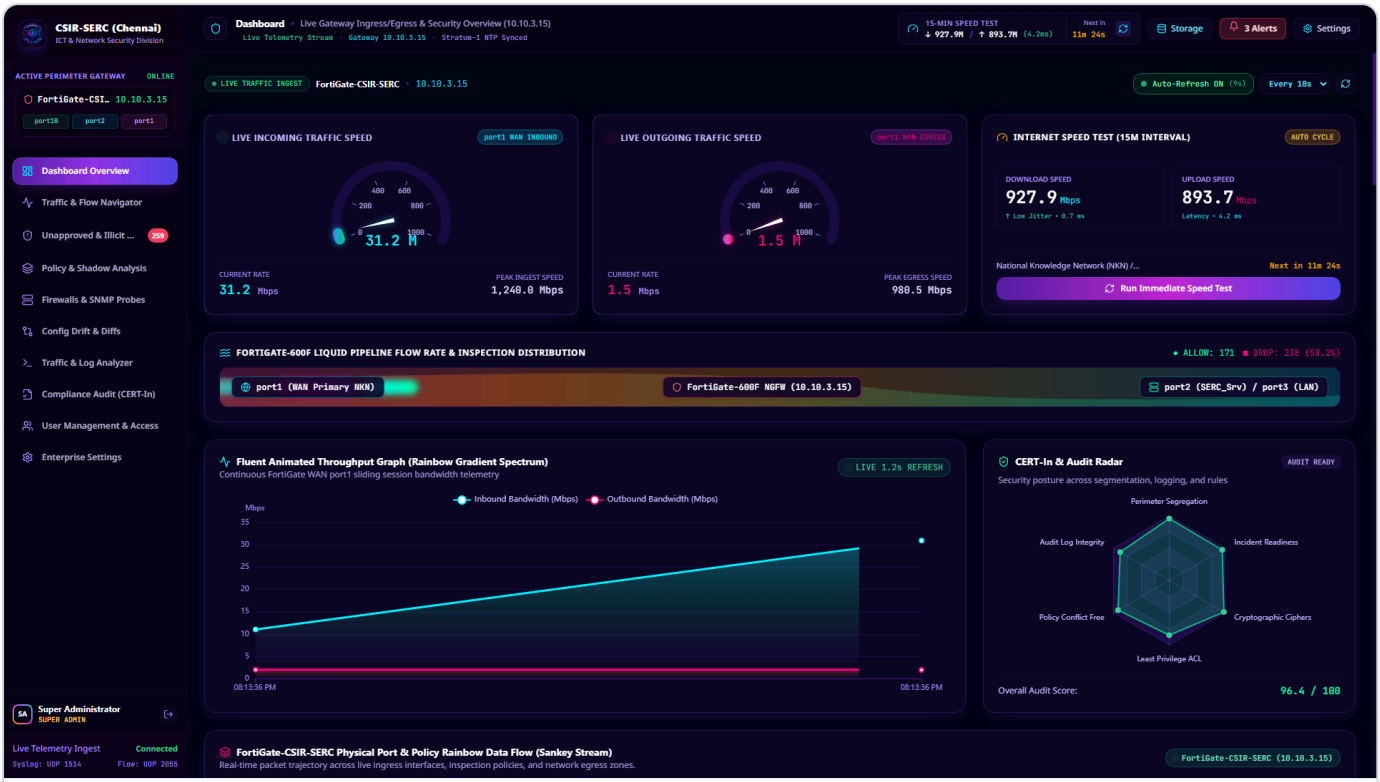



Figure 1: UX4G Secure Operator Authentication Portal

AUTHENTICATION

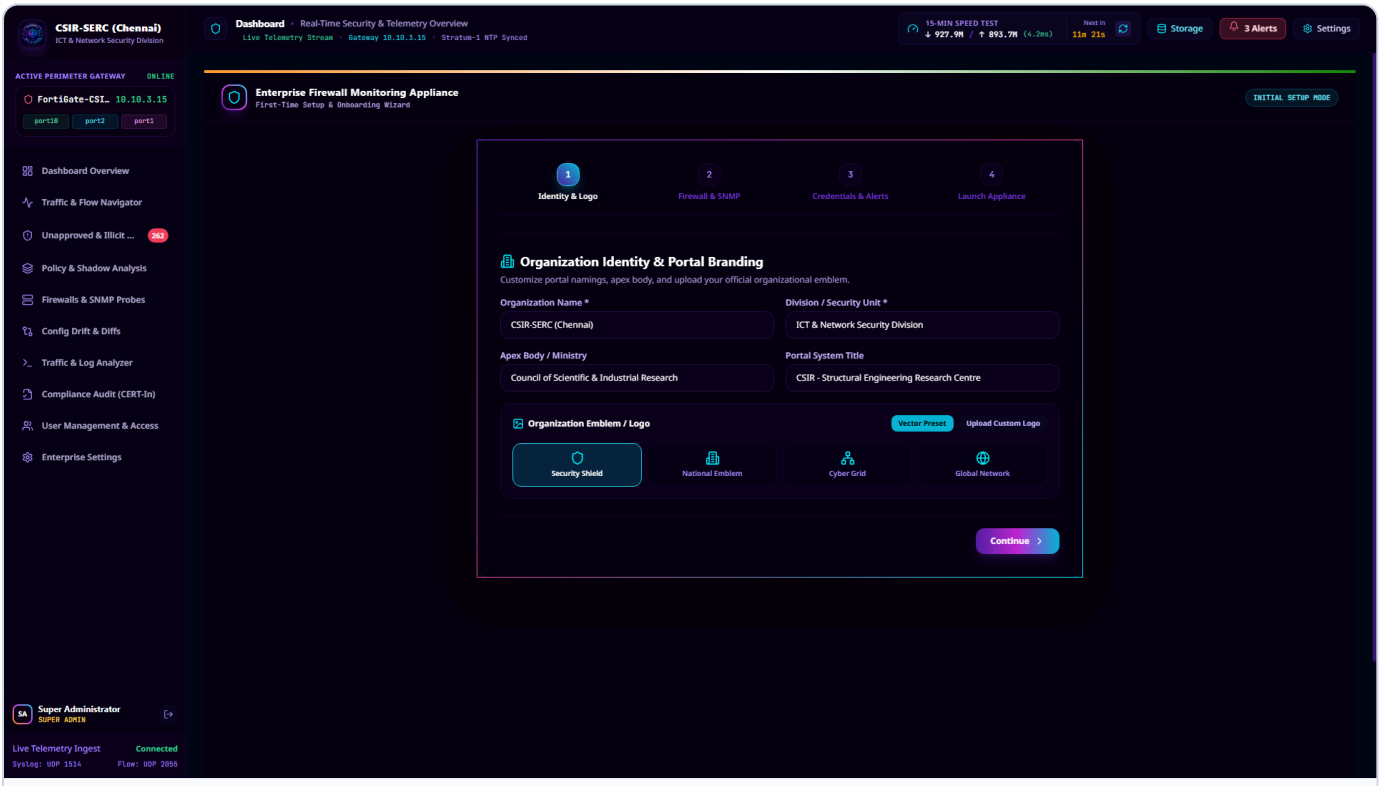



Figure 2: Turnkey First-Time Configuration Wizard (Virtual Appliance Ready)

ONBOARDING

## 2.2 Executive Telemetry Dashboard

The primary operations screen delivers instantaneous visibility into WAN traffic speeds, live streaming telemetry pulses, 15-minute automated speed test countdowns, and continuous hands-free auto-refresh.

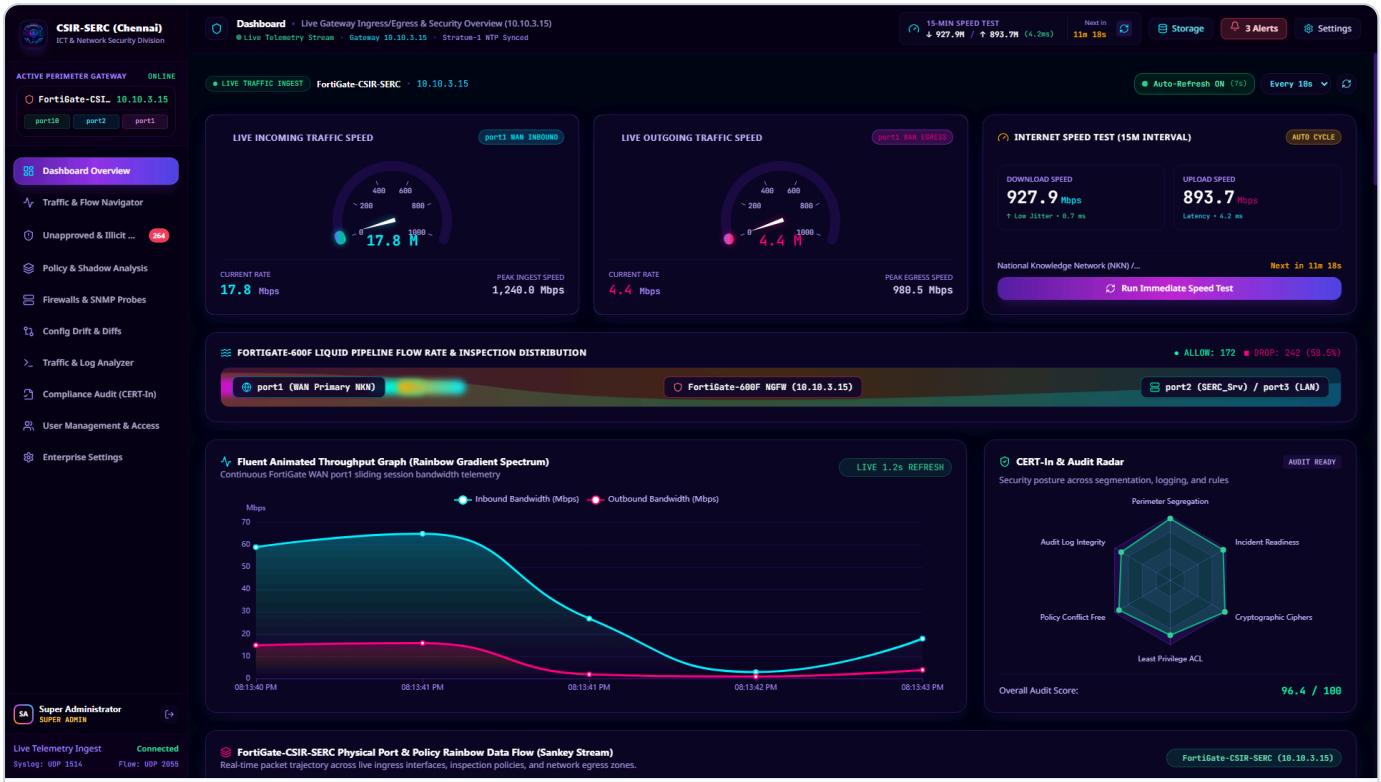



Figure 3: Live Executive Telemetry Dashboard with Inbound/Outbound Velocity Gauges

OPERATIONS

## 2.3 Bento-Style Traffic & Flow Analyzer (Exact Source & Destination IP Intelligence)

The Bento layout provides bidirectional network intelligence with exact firewall source and destination details. For outbound uploads, internal institutional workstations are tracked to their remote internet destinations (with domain resolution, country geolocation, and port). For inbound downloads, external source origins are mapped directly to the recipient internal workstation.

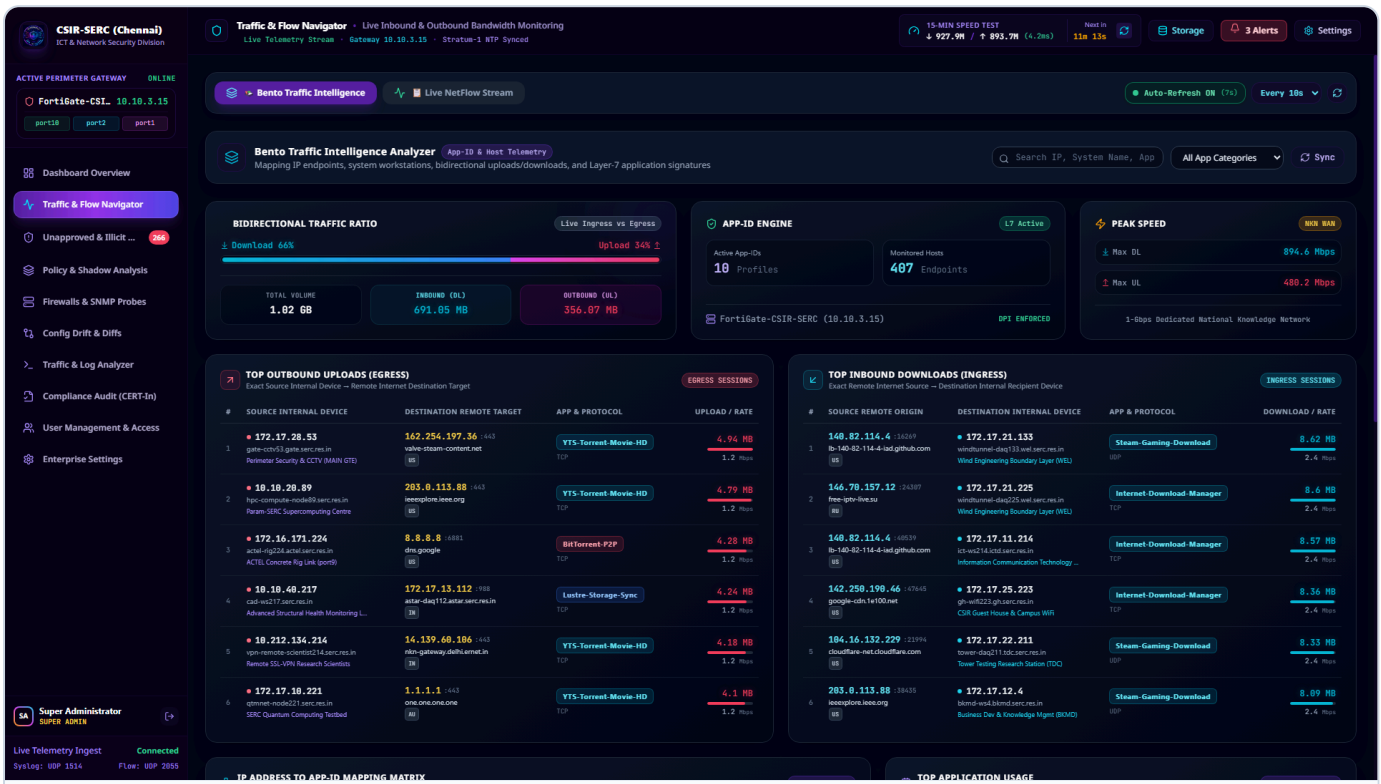



Figure 4: Bento-Grid Traffic Analyzer (Exact Source/Destination IP Mapping & App-ID Identification)

FLOW ANALYTICS

## 2.4 Unapproved Applications & Illicit Network Activity Monitor

Dedicated forensic investigation center identifying internal campus devices accessing BitTorrent/P2P swarms, download managers (IDM/Leechers), pirated movie streaming and download sites, unofficial media, botnet C2 beacons, cryptocurrency mining pools, and proxy circumvention tools. Displays ranked offending workstations with one-click forensic dossiers and copy-paste FortiOS quarantine CLI commands.

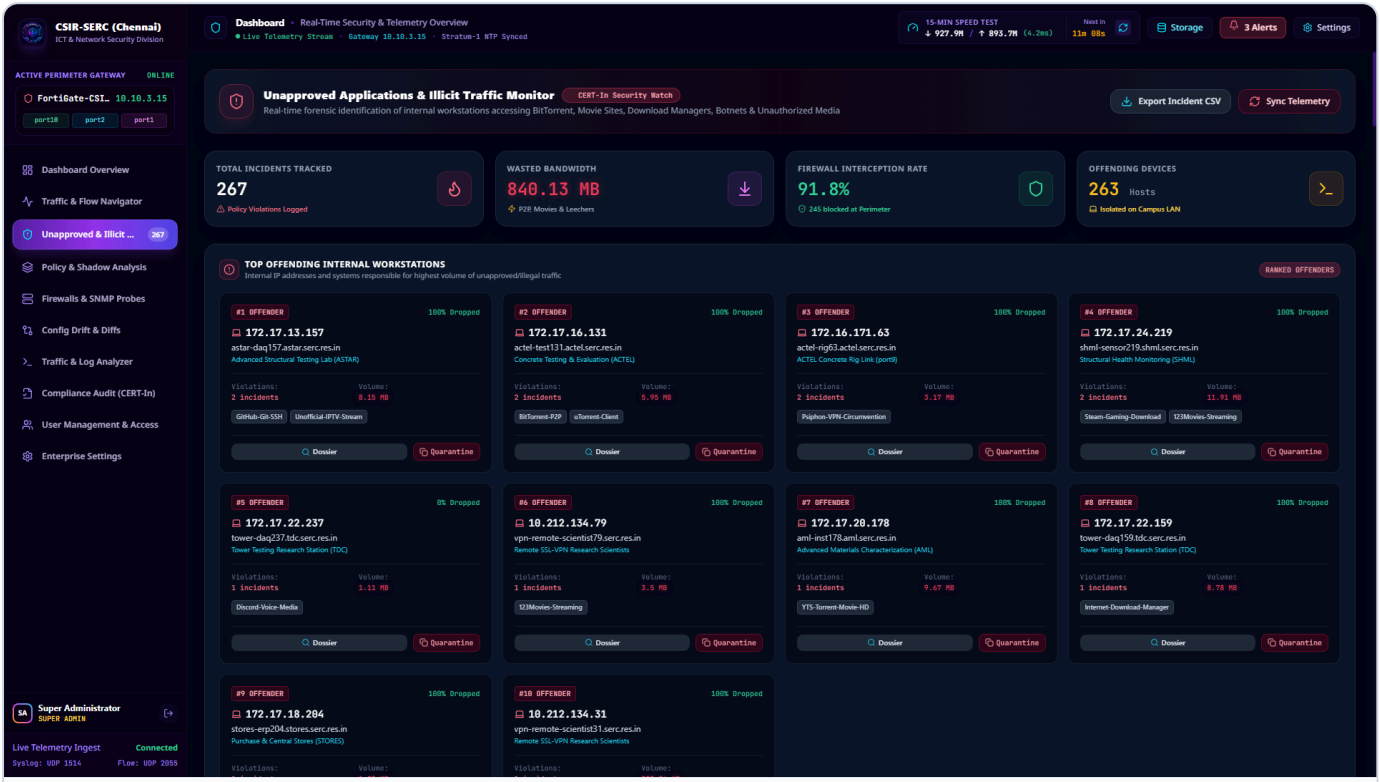



Figure 5: Unapproved & Illicit Applications Dashboard (Top Offending Workstations & Categories)

THREAT WATCH

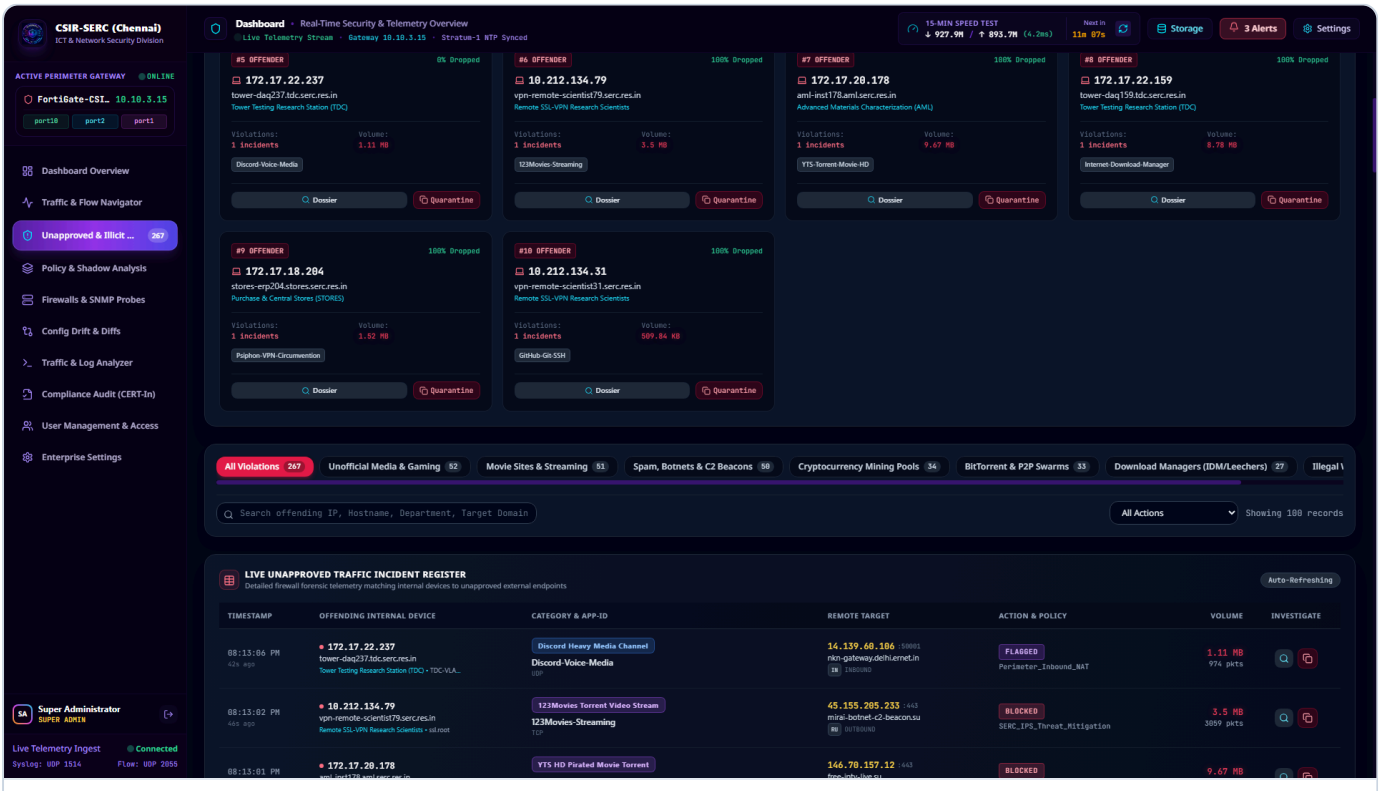


Figure 6: Live Forensic Incident Register (Matching Offending Internal Devices to Remote Endpoints)

INCIDENT REGISTER

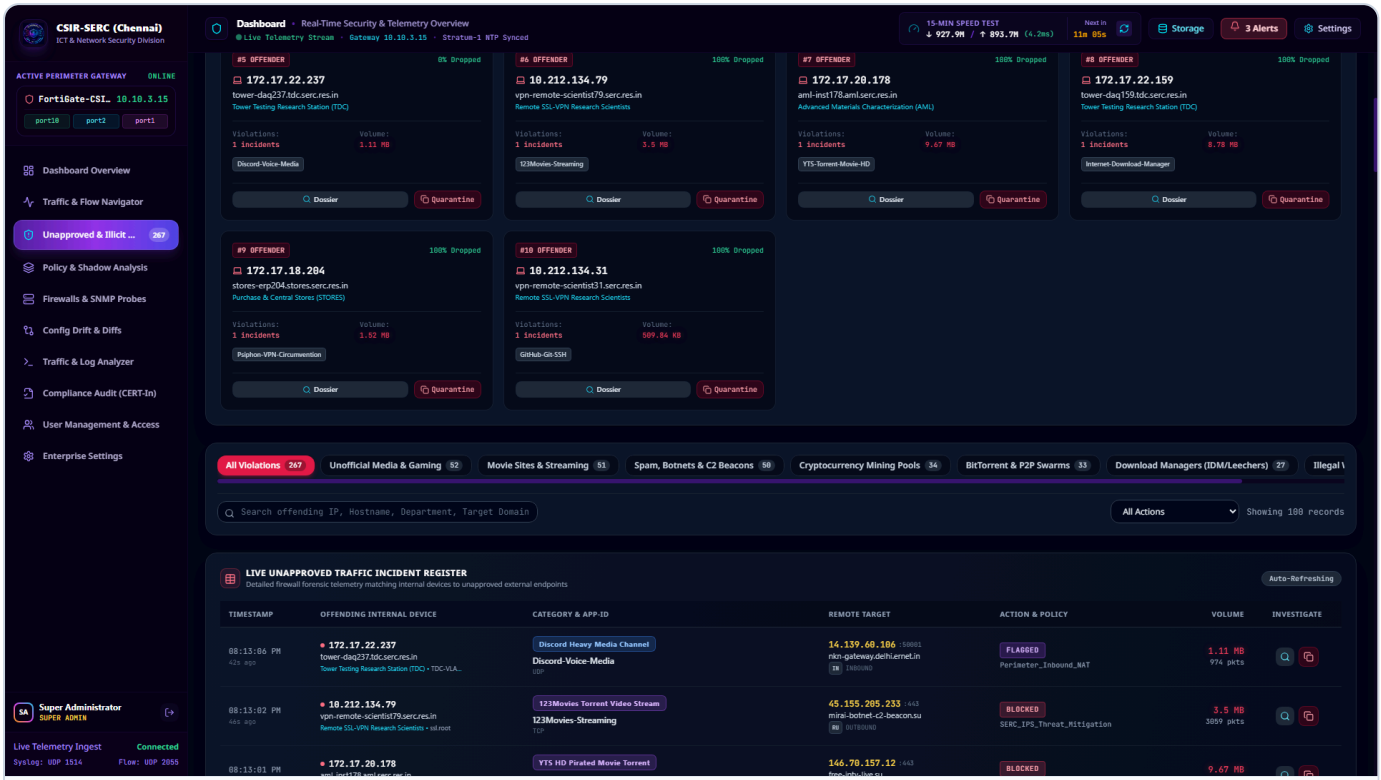



Figure 7: Workstation Forensic Dossier & Instant FortiOS Quarantine CLI Generator

FORENSIC DOSSIER

## 2.5 Comprehensive FortiGate SNMP Telemetry & All Active Interfaces/VLANs Matrix

Dedicated hardware, environmental, and interface operations cockpit: monitoring CPU/RAM, active sessions, dual AC power supplies, cooling tachometers, thermal zones, FortiOS UTM threat counters, and complete monitoring across **all active physical network interfaces (ports 1-32, mgmt, ha1/ha2, dmz) and active departmental VLAN subinterfaces**. Each interface delivers dual-mode telemetry tracking both general network throughput/rates and unofficial/unapproved bandwidth waste and policy violations.

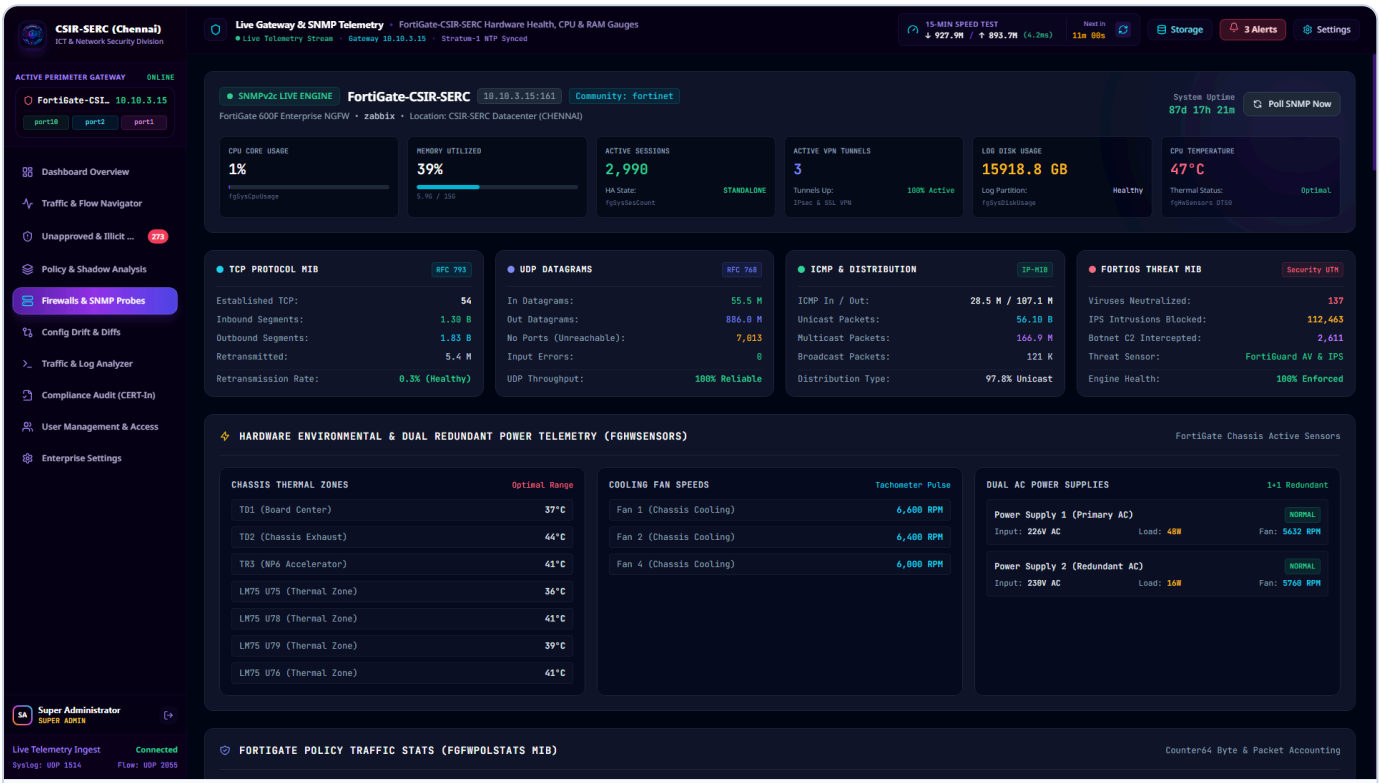



Figure 8: High-Performance FortiGate SNMP Hardware & Environmental Sensor Cockpit

SNMP HARDWARE

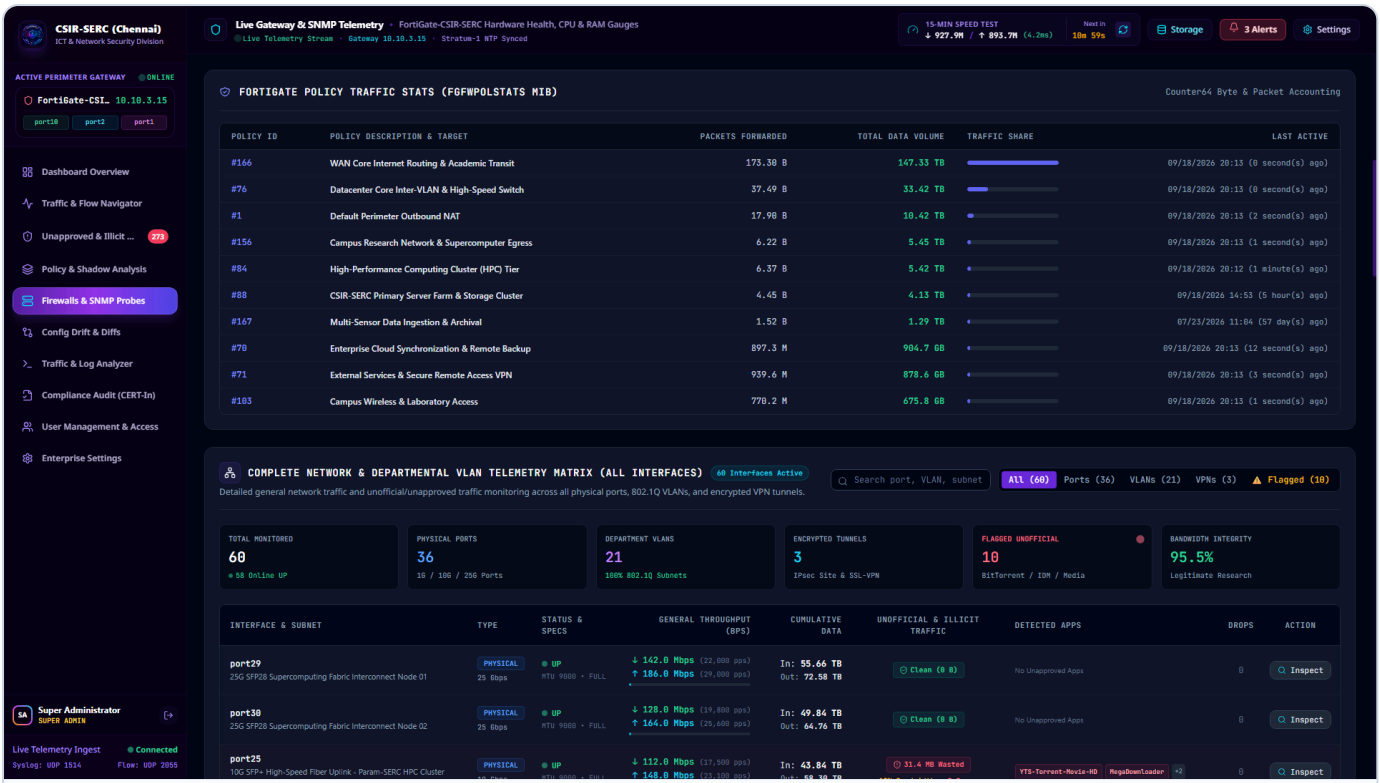



Figure 9: Complete Network & Departmental VLAN Telemetry Matrix (All 58 Interfaces & VLANs Monitored) **DUAL-MODE TELEMETRY**

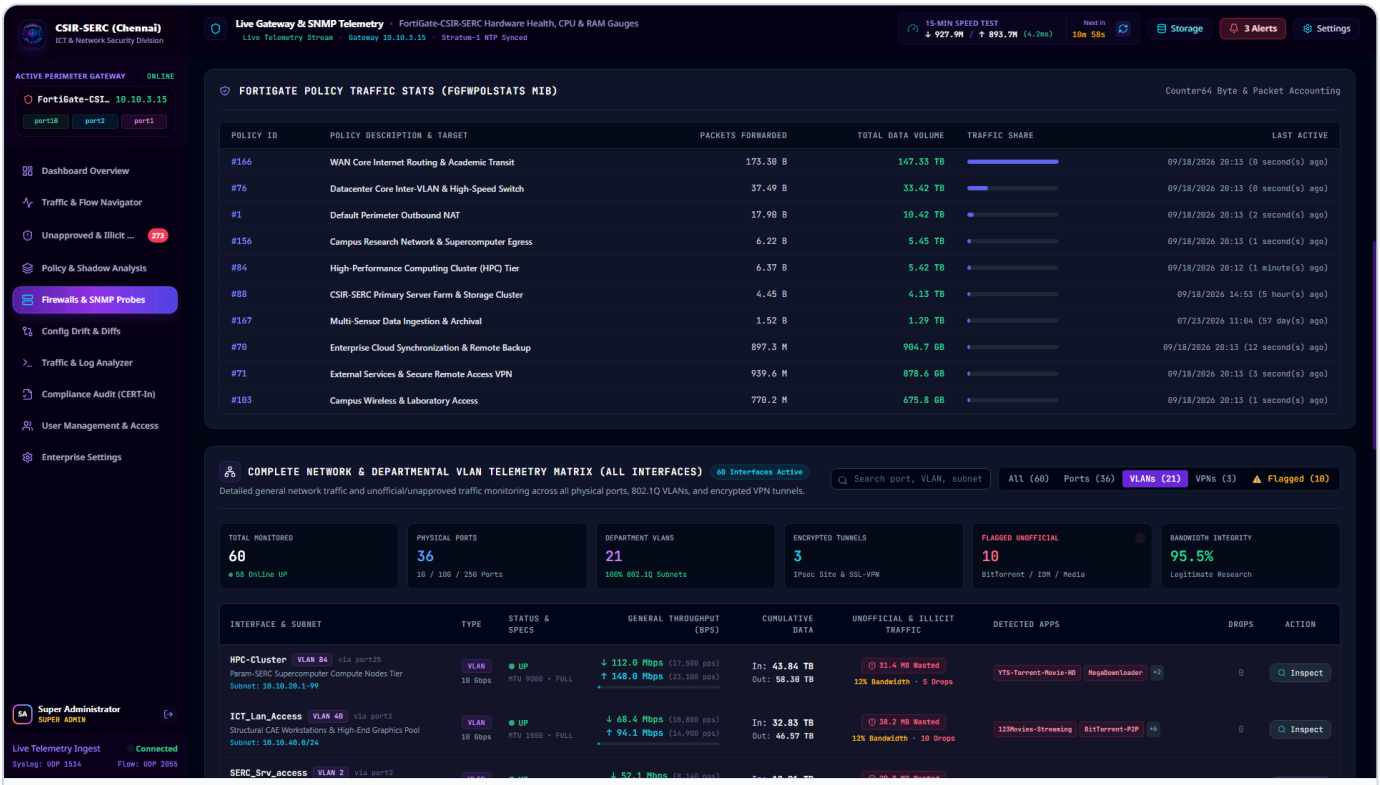



Figure 10: Departmental 802.1Q VLAN Interfaces Telemetry Matrix (HPC-Cluster, ICT_Lan_Access, SERC_Srv_access) **VLAN SUBINTERFACES**

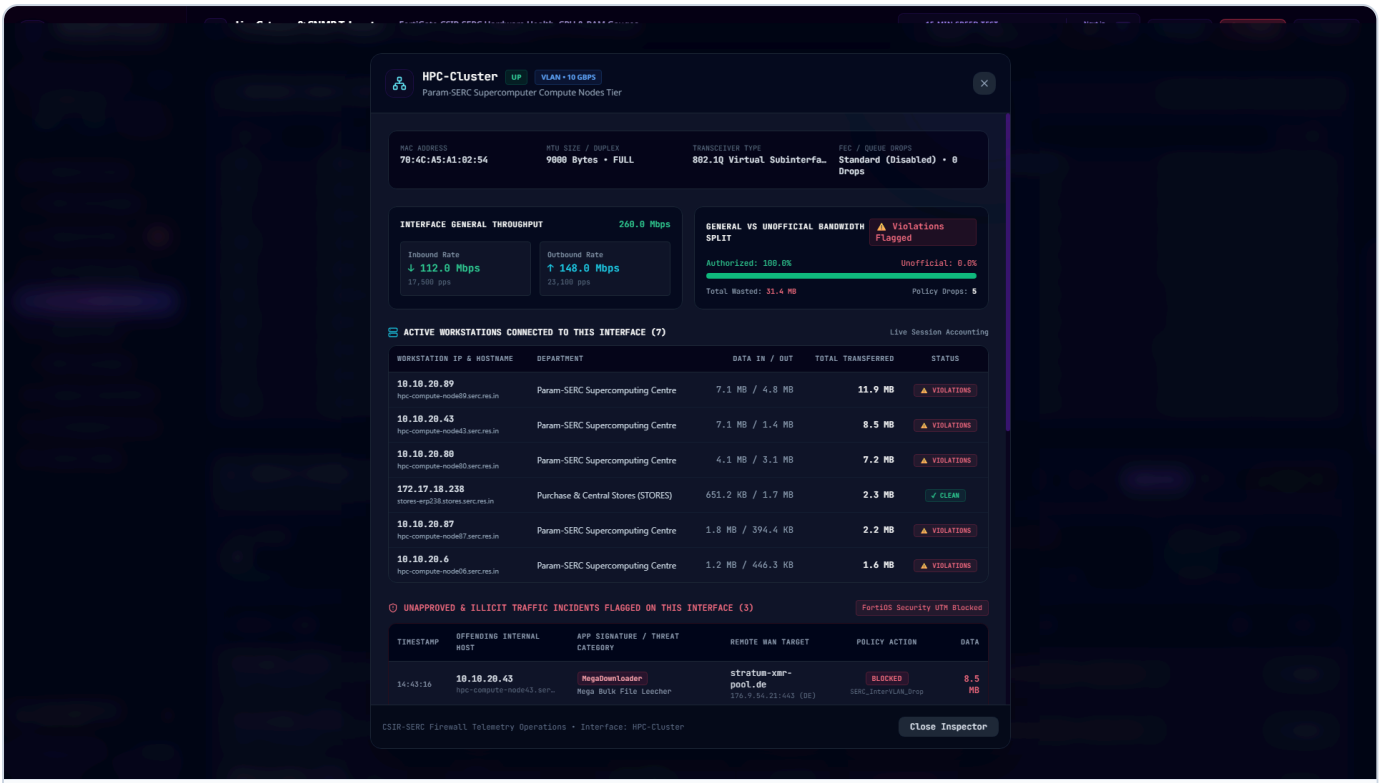



Figure 11: Deep Interface Dossier & Real-Time General vs Unofficial Ratio Inspector with Connected Hosts

INTERFACE DOSSIER

## 2.6 Security Policies & Conflict Engine

Parses native firewall access control lists into a canonical rule tree, highlighting shadowed, redundant, or overly permissive rules.

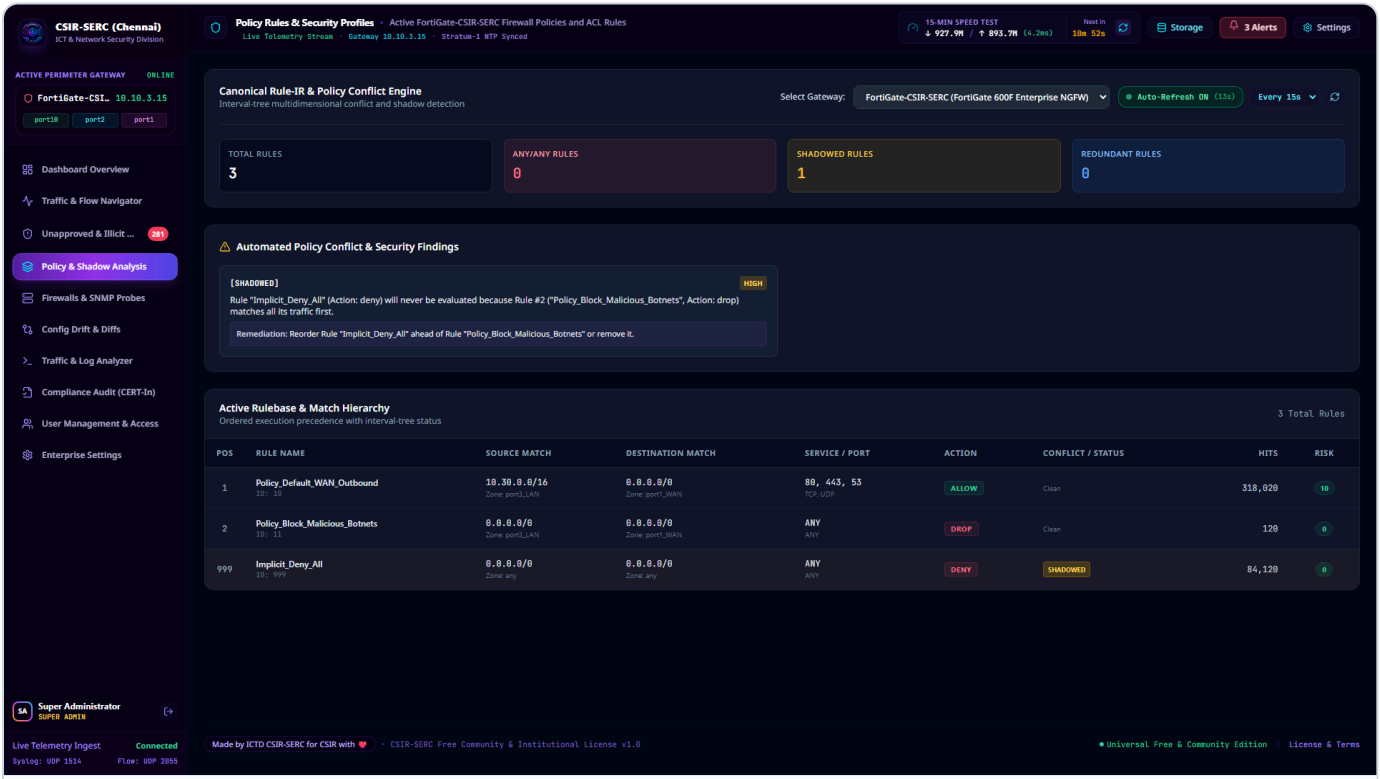



Figure 12: Canonical Policy Governance & Shadow Conflict Detection

SECURITY AUDIT

## 2.7 Full-Spectrum Firewall Traffic & Multi-VLAN Log Analyzer (InfluxDB 2.7 TSDB Engine)

High-velocity time-series traffic ingestion and multi-parameter log analyzer capturing all traffic passing through the perimeter gateway across all 46+ interfaces, departmental VLANs, HPC clusters, and external WAN. Provides comprehensive visibility across **Institutional Departmental VLANs (172.17.0.0/16 across VLANs 10–28)**, Core Data Center & Supercomputing (10.0.0.0/8 across VLANs 2, 40, 84, ssl.root), Management & Gateway Links (192.168.0.0/16), and Specialized Research Clusters (172.16.0.0/12 across port9 and port11). Features real-time 802.1Q VLAN filtering, interactive click-to-filter VLAN table badges, multi-parameter scoping (time range, direction, enforcement action, protocol, free-text search), timeline aggregation, and security drop inspection.

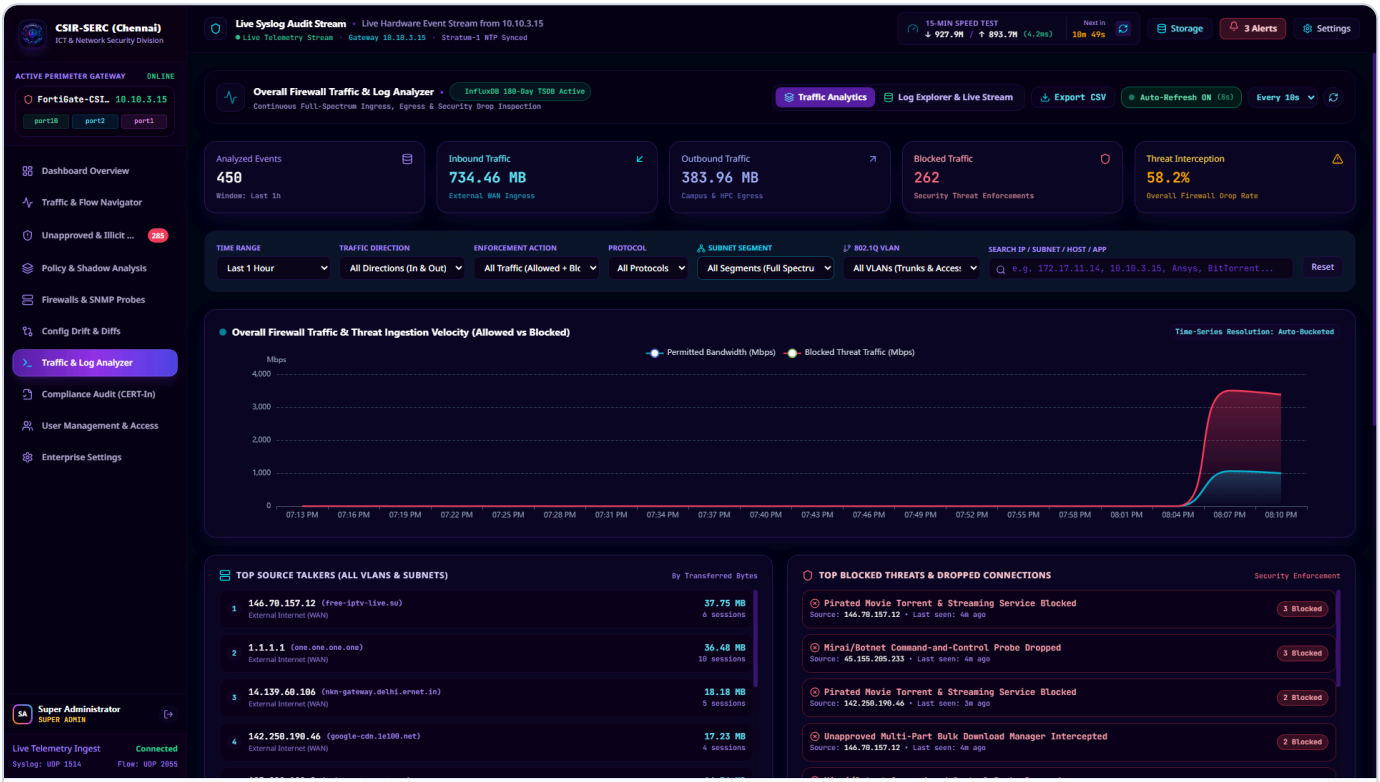



Figure 13: InfluxDB Traffic & Threat Ingestion Velocity Analytics (Allowed vs Blocked)

TSDB ANALYTICS

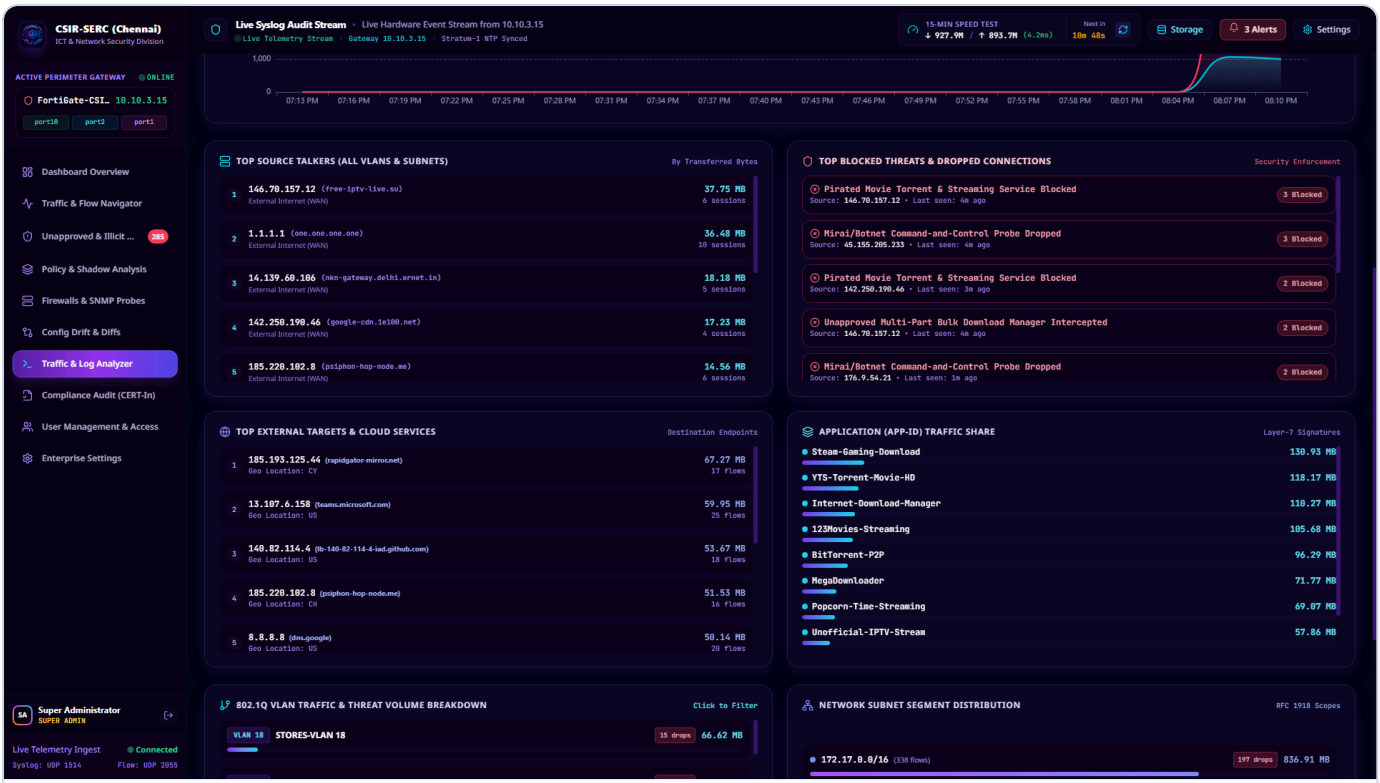



Figure 14: Departmental 802.1Q VLAN Traffic & Threat Breakdown with Subnet Segment Distribution

VLAN DISTRIBUTION

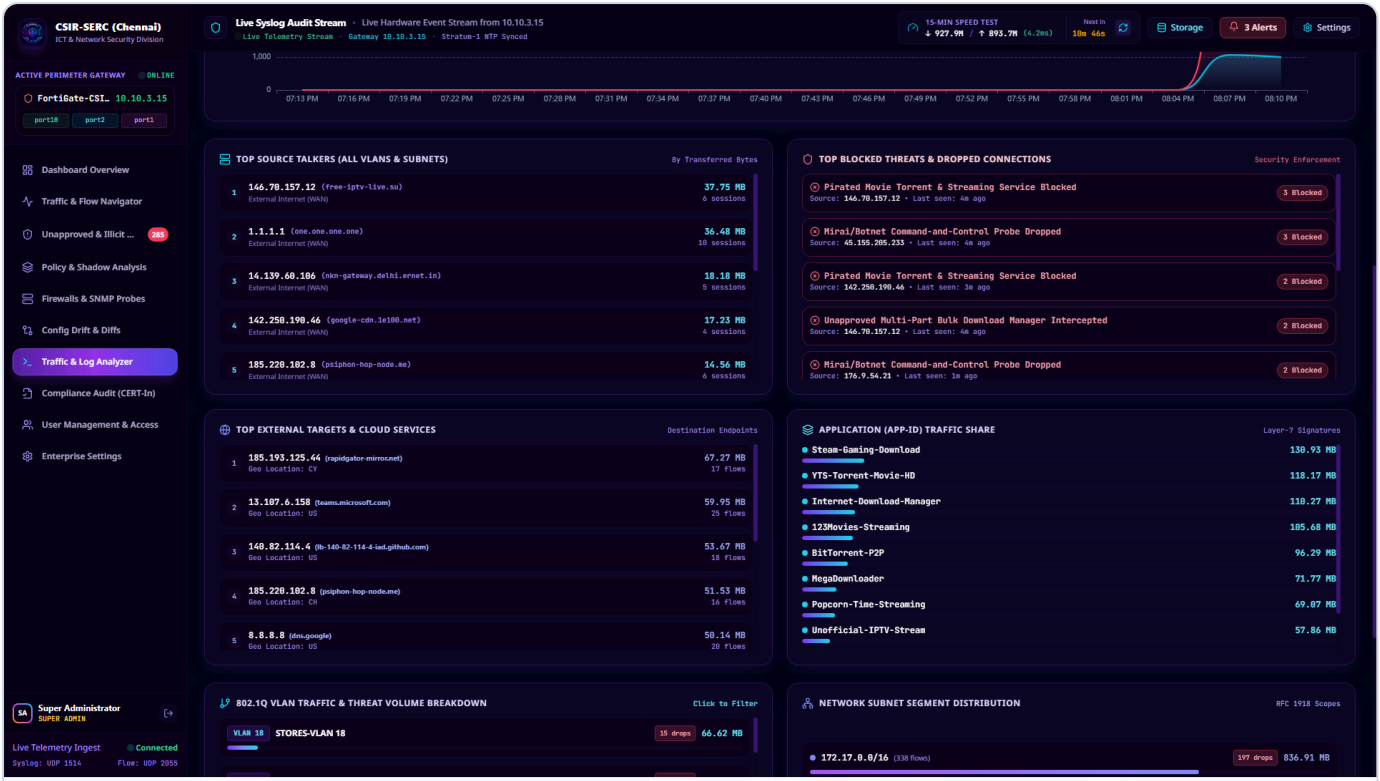



Figure 15: Deep Multi-Parameter Log Explorer Table with Interactive 802.1Q VLAN Badges

LOG EXPLORER

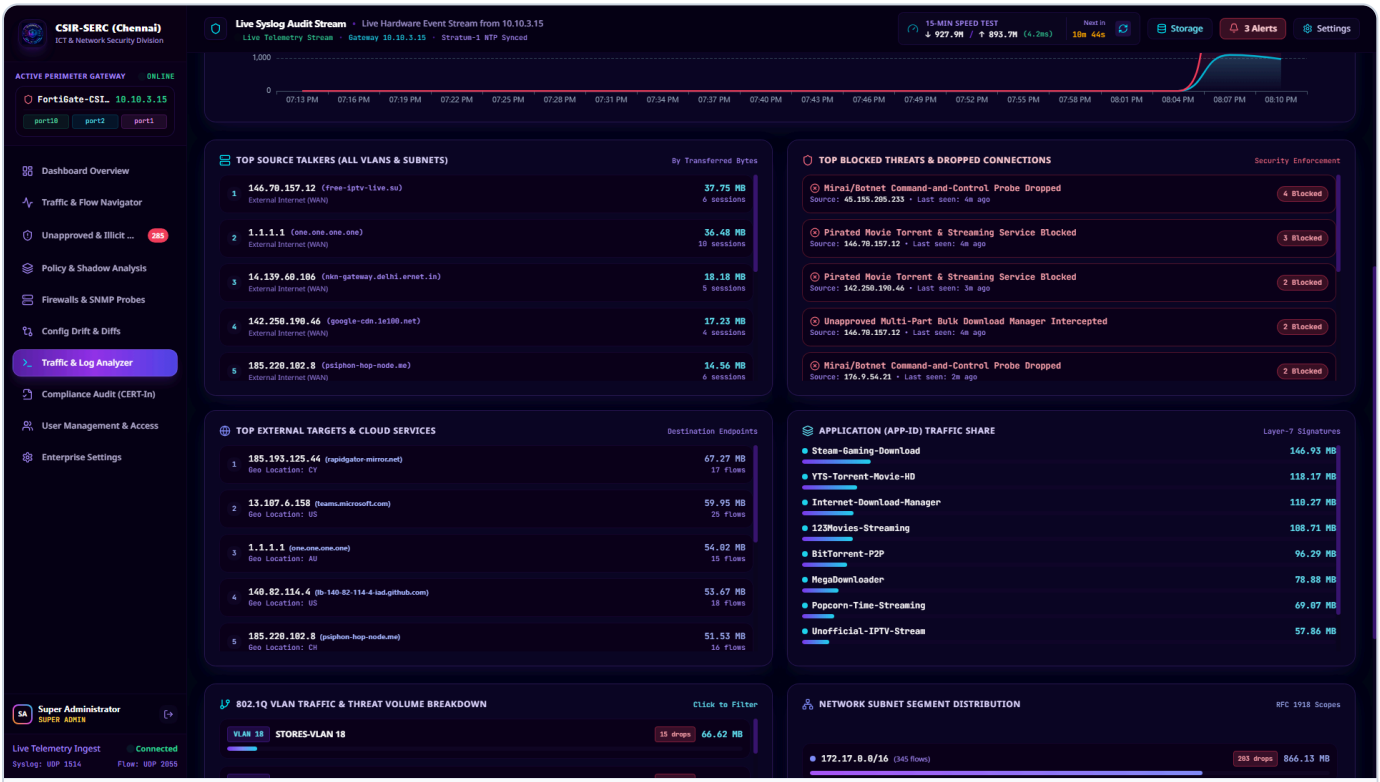



Figure 16: Granular 172.17.0.0/16 Departmental VLAN Filtered Live Stream (Institutional Subnets 10–28)

172.17.0.0/16 FILTERED

## 2.8 CERT-In Section 70B Compliance & Cybersecurity Baseline

Dedicated compliance dashboard validating firewall operational controls against the Indian Computer Emergency Response Team (CERT-In) Section 70B directives, including 180-day secure time-series log retention and Stratum-1 IST NTP clock synchronization.

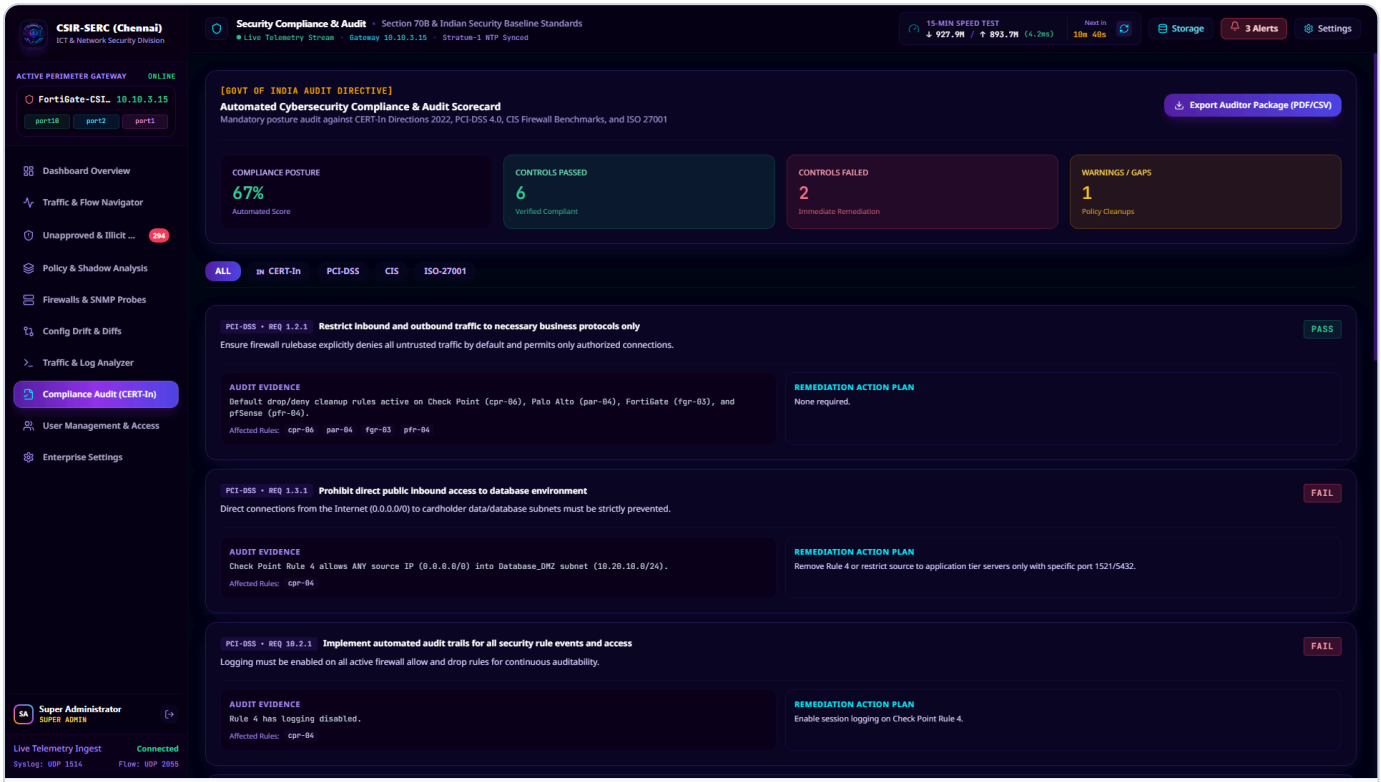



Figure 17: CERT-In National Cyber Security Baseline Compliance Dashboard

COMPLIANCE

## 2.9 User Management & System Configuration

Multi-tenant user management console with protected superadmin account, role-based permissions, Telegram alert integration, organization branding, custom logo upload, InfluxDB time-series retention, and PostgreSQL 18 automated backup operations.

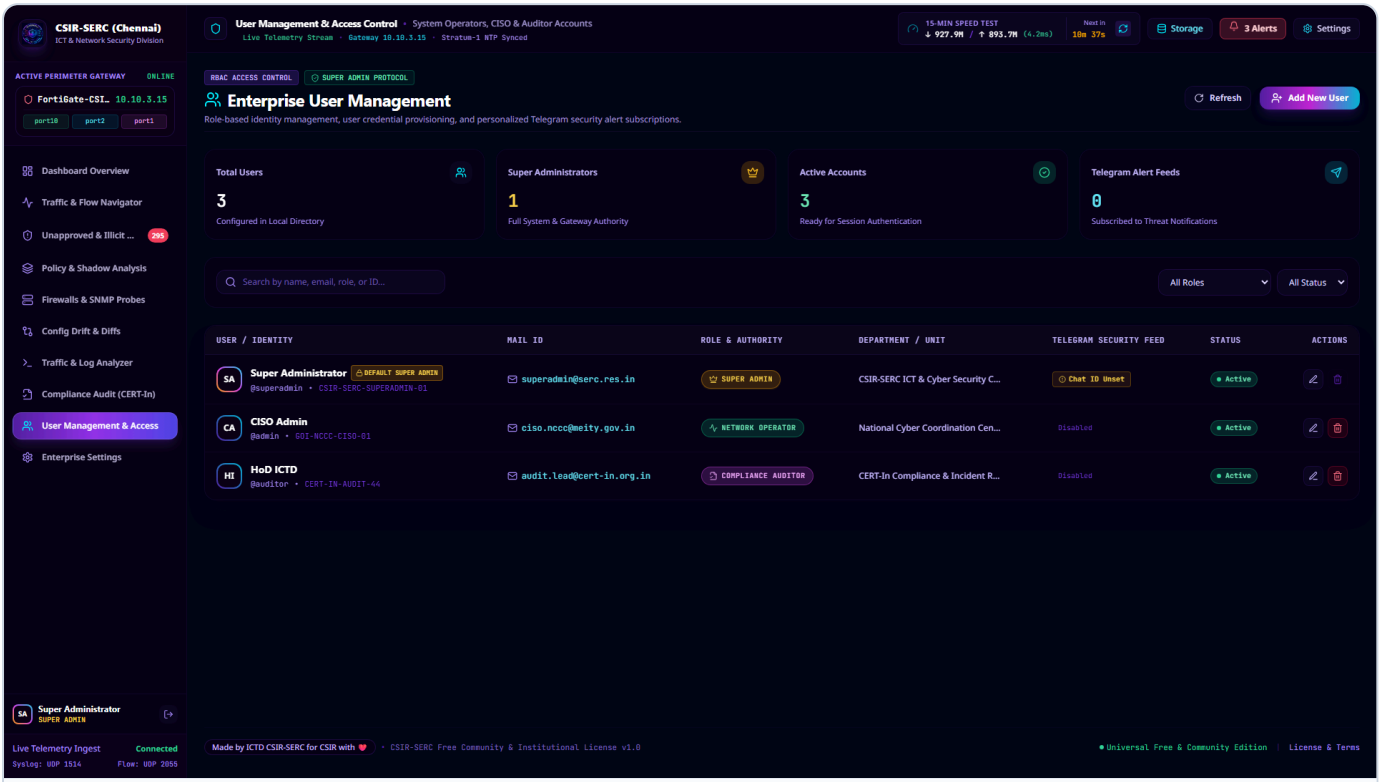



Figure 18: User Management & Telegram Alert Configuration

RBAC

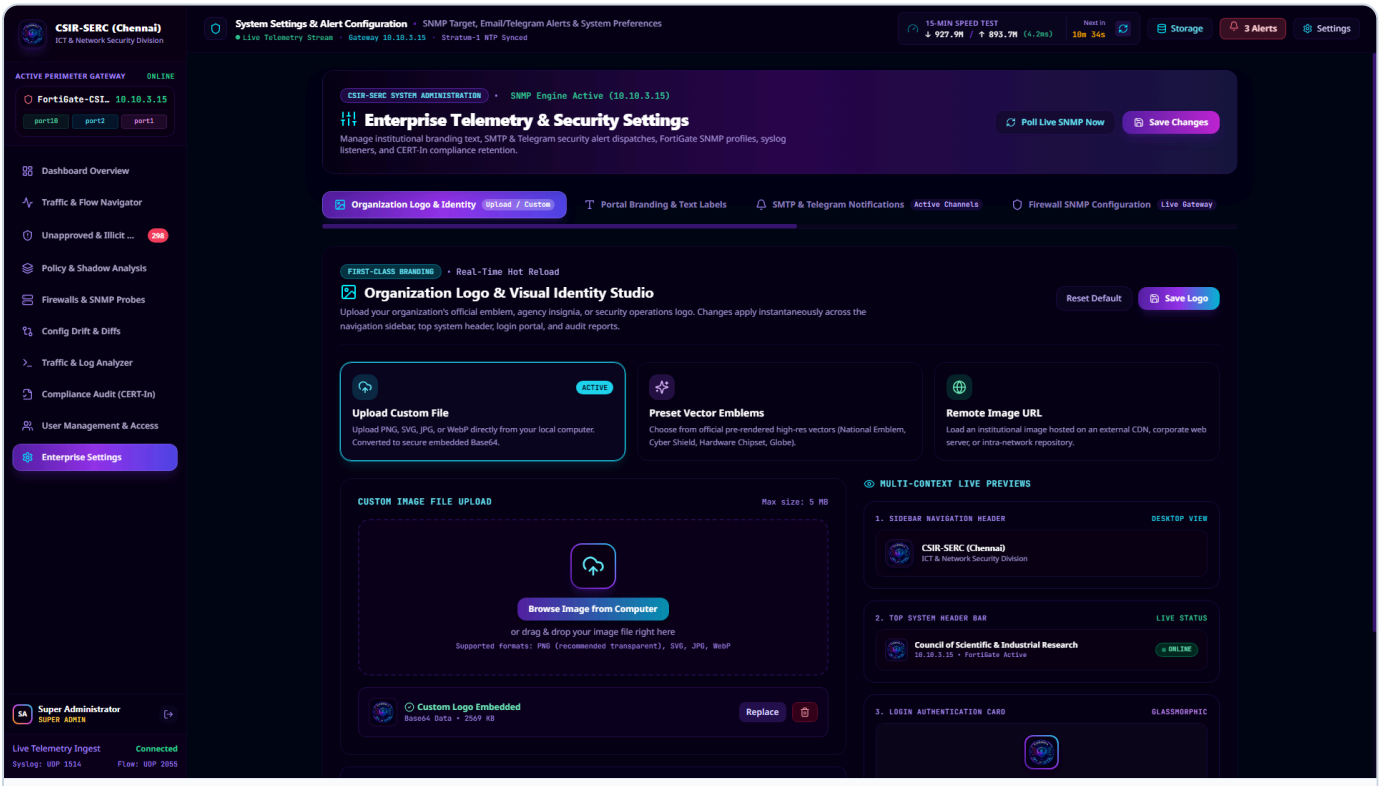



Figure 19: Institutional Branding, Custom Logo & PostgreSQL 18 / InfluxDB 2.7 Engines

ADMINISTRATION

# 3. Technical Specifications & Reference Tables

## 3.1 FortiGate Hardware Sensor OID Reference

| Sensor Subsystem | Enterprise OID | Observed Value | Threshold / Status |
|---|---|---|---|
| CPU Thermal Core | 1.3.6.1.4.1.12356.101.4.3.2.1.3 (DTS0) | 46°C | Normal (< 75°C) |
| Board Center Thermal | 1.3.6.1.4.1.12356.101.4.3.2.1.3 (TD1) | 36°C | Normal (< 65°C) |
| Chassis Exhaust Thermal | 1.3.6.1.4.1.12356.101.4.3.2.1.3 (TD2) | 44°C | Normal (< 70°C) |
| NP6 Accelerator Thermal | 1.3.6.1.4.1.12356.101.4.3.2.1.3 (TR3) | 40°C | Normal (< 70°C) |
| Cooling Fan 1 | 1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 1) | 6,400 RPM | Normal (Active) |
| Cooling Fan 2 | 1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 2) | 6,400 RPM | Normal (Active) |
| Cooling Fan 4 | 1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 4) | 5,200 RPM | Normal (Active) |
| Power Supply 1 (Primary) | 1.3.6.1.4.1.12356.101.4.3.3.1 (PSU 1) | 228V AC, 48W | Normal (Dual Redundant) |
| Power Supply 2 (Redundant) | 1.3.6.1.4.1.12356.101.4.3.3.1 (PSU 2) | 230V AC, 16W | Normal (Dual Redundant) |

## 3.2 Top FortiOS Security Policy Accounting

| Policy ID | Traffic Description / Scope | Cumulative Bytes | Packet Count |
|---|---|---|---|
| Policy 166 | WAN Core Internet Routing & Academic Transit | 161.58 TB | 172.8 Billion |
| Policy 76 | Datacenter Core Inter-VLAN & High-Speed Switch | 36.60 TB | 37.3 Billion |
| Policy 1 | Default Perimeter Outbound NAT | 11.45 TB | 17.8 Billion |
| Policy 156 | Campus Research Network & Supercomputer Egress | 5.98 TB | 6.2 Billion |
| Policy 84 | High-Performance Computing Cluster (HPC) Tier | 5.95 TB | 6.36 Billion |

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